北京大学
PEKING UNIVERSITY

Beyond OCR: A Survey of Document Intelligence from Structured Perception to Multimodal Question Answering

JiaShu Yang[1] **Chi Zhang**[1] **Xingyu Qi**[1] **Ning Zhang**[1] **Jie Wu**[1] **Lingxu Chen**[1] **Jiani Guo**[1]

[1]Peking University

Abstract

Document intelligence connects page perception with reasoning over text, layout, tables and images. Its central challenge is preserving the evidence needed for an answer across recognition, representation, retrieval and computation. This survey compares OCR-based, direct-visual and hybrid approaches through five questions: what representation is used, how much context is accessed, which evidence units are retained, how inference is controlled, and what the evaluation verifies. Visual worked examples connect these choices to concrete errors, including lost headers, missing cross-page corrections, insufficient image resolution and unsupported citations. We organize representative methods by their mechanisms rather than list papers chronologically, and distinguish answer correctness, evidence localization, source support and execution. The resulting synthesis guides the selection of document-QA workflows under explicit input and resource constraints.

Project Page: https://yang-jiashu.github.io/document-intelligence-survey/

Code Repository: https://github.com/Yang-Jiashu/document-intelligence-survey

Correspondence: jiashuyang827@gmail.com

1 Introduction

Document intelligence must connect what a page shows with what an answer claims. A financial question may require a table cell, its year and unit, and a correction on another page. An answer can fail at any link: character recognition, row–column alignment, page retrieval or calculation. Figure 1 follows these links from a document to a source-backed answer.

1.1 Representations and Evidence

OCR-based systems expose text and coordinates; visual models encode page appearance; hybrid systems combine both. These interfaces preserve different information. OCR may miss a symbol or its reading order, while image resizing may erase fine print. Retrieval decides which evidence reaches the reader, and tools can make numerical operations explicit. A system may combine all of these mechanisms.

We compare methods through five questions: **representation** (text, coordinates or pixels), **context** (region, page or collection), **granularity** (token, cell or page), **control** (fixed or iterative access), and **validation** (answer,

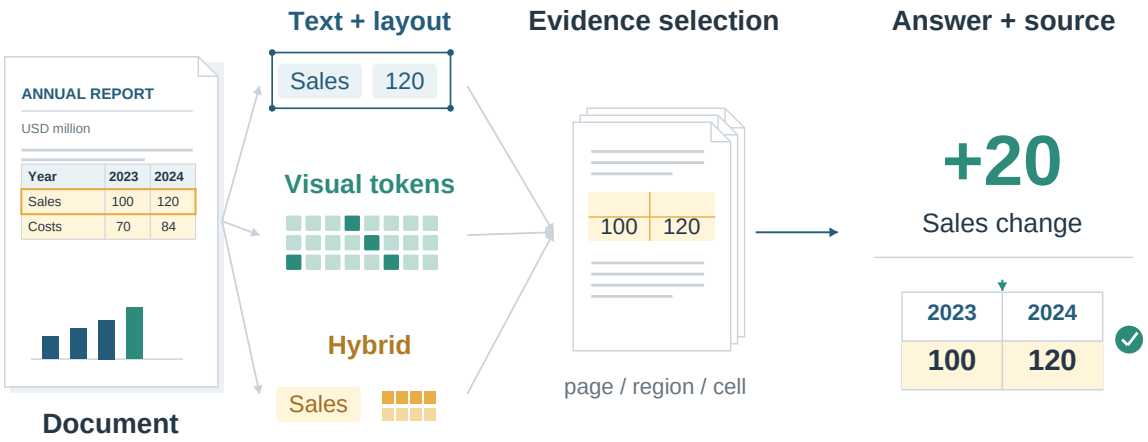


Figure 1 From a page to a source-backed answer. Text/layout, visual and hybrid representations feed evidence selection; the constructed sales example keeps the answer linked to its source cells.

source or computation). Table understanding is a task family, RAG an access strategy and a VLM a model family; they are compatible choices rather than competing categories.

One question, two pages. Figure 2 shows the running example. A table gives margins of 15% and 12%, but a separate note restates the earlier profit from 96 to 104. With revenue unchanged at 800, the corrected margin is 13%: the increase is 2 percentage points. Retrieving the table alone gives a plausible but incorrect 3-point answer. Executing subtraction cannot repair the missing note.

1.2 Scope and Reading Map

This curated narrative review covers document layout, OCR, table reasoning, retrieval, visual models and evaluation. Existing surveys organize LLM-based document AI [73] and multimodal document RAG [19]; our focus is how representations and inference choices preserve, select and verify evidence. Representative mechanisms appear in the main text, with the broader literature retained in the project's searchable reading catalog.

Figure 3 provides a chronological entry point. Dates refer to first public papers; yellow underlines mark official trained weights verified on 9 November 2026. The chronology records coexistence rather than a ranking or replacement of earlier methods. This survey synthesizes published designs and protocols without introducing a new model benchmark.

Sections 2 and 3 examine page and table representations; Sections 4 and 5 compare evidence access and visual processing. Section 6 separates answer, evidence and execution checks, and Section 7 identifies research priorities.

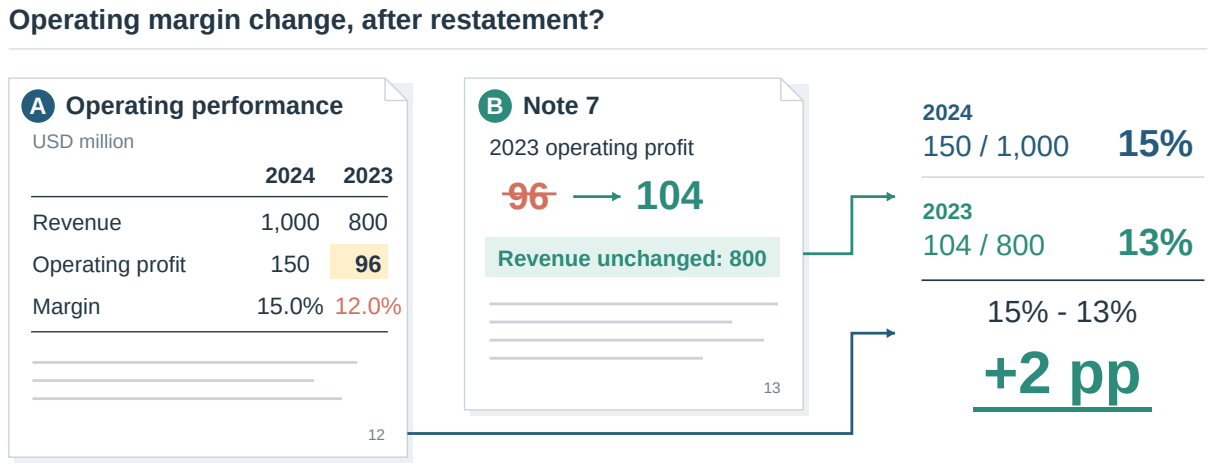


Figure 2 Constructed cross-page example: table A and restatement B jointly support $150/1000 - 104/800 = 0.02$, a 2-percentage-point increase. Colored links preserve the source cells and correction used in the calculation.

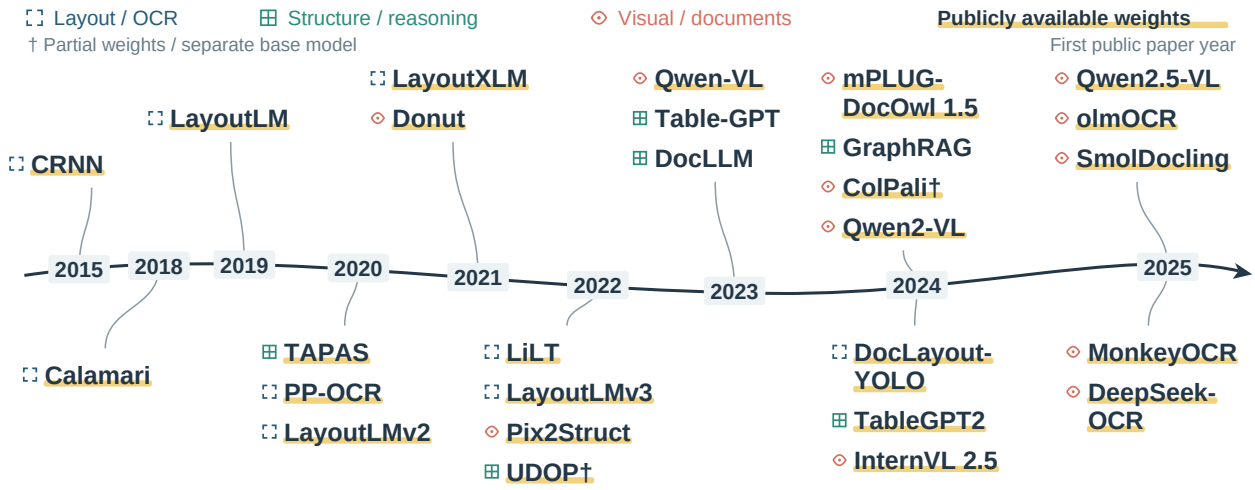


Figure 3 Selected document-AI papers by first public year. Yellow underlines mark verified official trained weights; daggers indicate partial/component releases. Names link to papers and underlines to weights; chronology does not imply model ancestry.

2 Document Layout Analysis and OCR

Perception constructs the textual and spatial evidence consumed by downstream document reasoning. Layout analysis and OCR are complementary interfaces, and their outputs may also be combined with direct visual representations.

2.1 Document Layout Analysis

A page is not a bag of words. The same number may be a table entry, a footnote marker or a page number; its role depends on position and relations to other content. Layout analysis makes those relations available to recognition, retrieval and reasoning. Three outputs should be distinguished: *regions* locate content, *relations* connect it, and *representations* encode both for a downstream model.

2.1.1 From Regions to Document Structure

A region detector predicts boxes or masks for text, titles, tables and figures. This provides useful processing units: a recognizer can receive a text crop, a table parser can receive a table region, and a retriever can retain the originating page. Detection is therefore an interface between page pixels and specialist components, rather than a complete account of document meaning.

The main design choice is how much context the detector sees. Local appearance helps distinguish a table from prose, but can be ambiguous when captions, columns and figures are tightly packed. VSR combines visual, semantic and relational features [91]; DocLayout-YOLO uses synthetic document data and global-to-local perception [93]. These approaches illustrate complementary ways to use context without assuming that all document types share the same layout.

Boxes alone leave important structure unresolved. A two-column page needs reading order; a table needs header–cell links; a figure needs its caption. Recovering these relations can be more consequential for QA than a small improvement in box overlap. For example, reading across columns may create a fluent but nonexistent sentence even when every word is recognized correctly.

Consider a report table whose unit appears above the box and whose qualification appears in a footnote below it. Cropping the cells accurately still removes context needed to interpret their values. A useful structural output therefore includes the table's neighborhood and links to its notes. The region boundary and the boundary of sufficient evidence are not necessarily the same.

2.1.2 Encoding Text and Spatial Relations

Layout-aware models pass OCR text and geometry into a learned representation. LayoutLM adds two-dimensional token positions to Transformer inputs [82]. Compared with plain text serialization, the model retains information about which words occur on the same line or near the same field. It still depends on the recognizer: an omitted word is absent from the token input, and an incorrect box changes the spatial evidence.

The representation can separate linguistic and geometric processing. LiLT uses a language-independent layout Transformer that interacts with text features [69], making layout transfer a different problem from learning a particular language. This is useful when document templates recur across languages, although script-specific recognition quality remains a separate dependency.

Reading order deserves explicit treatment. XYLayoutLM addresses ordering noise through an augmented XY-cut procedure and positional encoding [22]; relation-based work instead represents order as directed links [88]. A sequence is convenient for a language model, while a relation graph can preserve alternatives and hierarchy. The appropriate interface depends on whether the question requires a passage, a table neighborhood or a cross-page link.

2.1.3 Using Structure During Reasoning

Structure can condition generation directly. DocLLM uses text and bounding boxes with disentangled attention, without an image encoder [68]. LayoutLLM introduces layout instruction tuning [43]. Both make

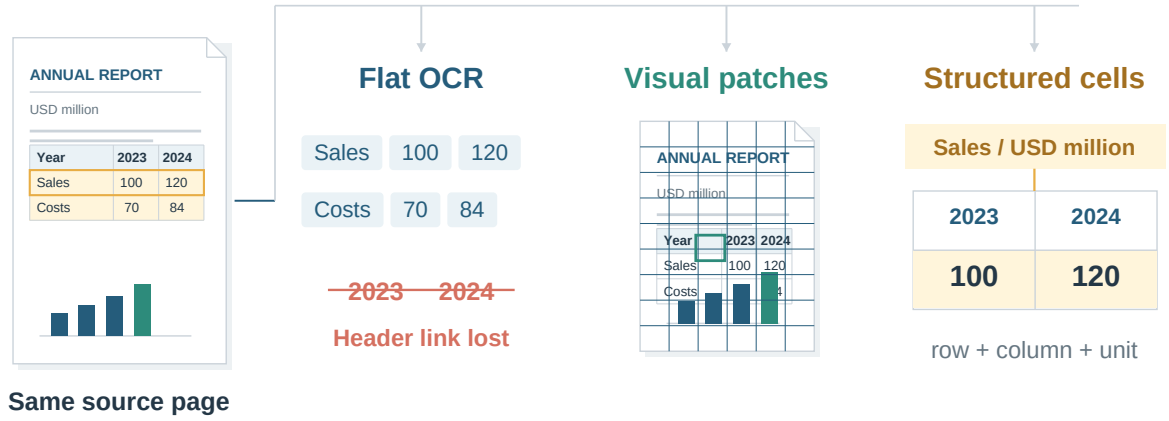


Figure 4 Three representations of the same page: flat OCR, visual patches and structured cells. The constructed OCR failure loses the association between values and year headers; the other panels show information that can be retained.

geometry available to the answering process; neither should be described as observing all visual content simply because spatial information is present.

Structure can also organize evidence access. LAD-RAG uses layout graphs for cross-page retrieval [61]. Here the benefit is not a new text recognizer but a different retrieval unit: a table, its caption and a linked paragraph can remain connected rather than become unrelated chunks. This is particularly relevant when a question needs both a numerical cell and an explanation elsewhere in the document.

These choices expose different failure points. Coordinate-aware generation can fail after a bad OCR pass; graph retrieval can fail when a required link was never built. The useful question is therefore which relations survive each interface. Real-document analyses show that performance of pretrained text–layout models can degrade outside their familiar distributions [89]; visual plausibility of the detected page is insufficient evidence of robust reasoning.

2.1.4 What Layout Benchmarks Measure

Dataset construction shapes the task. PubLayNet obtains large-scale region supervision by aligning scientific PDFs with XML [94]; DocBank provides token-level document annotations [35]; DocLayNet uses human-annotated pages from several document domains [57]. These resources offer different balances of scale, label granularity and domain diversity.

Region overlap and category accuracy measure detection, while reading-order and relation annotations test structure. Downstream QA adds another requirement: the selected regions must jointly support an answer. Evaluations should connect these levels by inspecting whether an error arose in the box, its semantic role, its links or the answer that used it. A high detection score cannot resolve those questions by itself.

2.2 OCR

OCR converts page appearance into content that other components can search or manipulate. The practical design space is broader than recognizing a string: a system may return words with coordinates, a serialized table, a structured page or a direct answer. Figure 4 contrasts the information exposed by these interfaces.

2.2.1 Detection and Sequence Recognition

A modular OCR pipeline separates page segmentation, text localization and recognition. OCRopus illustrates the combination of layout processing with learned recognition [3]; Calamari uses convolutional and recurrent

sequence modeling with CTC training [79]. CTC allows training from line transcripts without requiring a character boundary for every symbol, while detection supplies the regions on which recognition operates.

This separation makes errors inspectable. A missed line is a detection error; a substituted digit is a recognition error; a correct word placed in the wrong table cell is a structural assignment error. Repairing one does not automatically repair the others. Returning coordinates and page identifiers with each transcript keeps these distinctions visible to later stages.

Deployment also changes the design objective. PP-OCR optimizes a compact detection-and-recognition system [13], while PaddleOCR 3.0 packages recognition, document parsing and information extraction into a broader toolkit [11]. A small recognizer may be sufficient for clean printed forms, whereas degraded scans or mixed formulas and tables can require different preprocessing and region-specific models. Architecture labels alone do not determine that trade-off.

2.2.2 Image-to-Sequence and Vision-Language Parsing

Generative document models decode visual features into task outputs. Donut uses an image-to-sequence interface without an external OCR transcript at inference [30]. VISTA-OCR uses a lightweight CNN encoder and an mBART-based Transformer decoder, initialized from Donut, to autoregressively emit text and quantized coordinates [25]. Prompted transcription and localization are therefore changes to the output interface, rather than evidence that localization is unnecessary.

Vision-language OCR couples image encoding to a language model. Ocean-OCR combines a dynamic-resolution NaViT-style encoder, an MLP projector and Qwen2.5-3B [5]. GOT addresses several content types through a unified OCR model [77]; dots.ocr jointly targets layout, recognition and relations [38]. These systems can serialize heterogeneous pages, but the resulting text must still retain the distinction between a heading, a table cell and a formula.

Visual-token allocation is a central constraint. Higher resolution exposes small characters but increases the amount of visual content to encode. DeepSeek-OCR studies optical context compression [78]; TextHawk2 addresses fine-grained perception and grounding with compressed visual representations [87]. Compression should be evaluated against the information needed by the task: a page summary may tolerate details that numerical QA cannot lose.

"OCR-free" refers here to the absence of an external transcript at the task interface. It does not mean that the model avoids learning to read, that OCR was absent from data construction or that generated answers are grounded automatically. In the original MMLongBench-Doc experiments, most tested LVLMs underperformed the corresponding OCR-text-based LLM setting [46]. That result is tied to the tested models and access conditions; it supports comparing input routes explicitly rather than treating direct visual input as a universal successor.

2.2.3 Hybrid Parsing and Targeted Re-observation

Hybrid systems allocate different parts of a page to different components. MonkeyOCR organizes parsing around structure, recognition and relations [40]; OCR-R1 interleaves a VLM's observations with specialist OCR tools [4]. A difficult formula can therefore receive different treatment from a clean paragraph. The price is an additional coordination problem: outputs from several components must agree on regions, reading order and the source of each extracted value.

The interface matters as much as the component list. A pipeline that keeps text, crops and coordinates can revisit an ambiguous digit; one that retains only plain text cannot. Conflicting tool outputs should remain associated with their source region so that a later check can compare them with the page. Calling more tools is useful only when the new observation can resolve a specific uncertainty.

OCR features can also support a task other than transcription. DocParseNet fuses visual features with OCR-derived text embeddings for semantic annotation of scanned agreements [50]. Its output identifies document regions and fields. This differs from reading every character or recovering a table's cell structure, so segmentation results should not be presented as recognition accuracy.

Route	Evidence passed forward	Useful control	Main failure point
Modular OCR	Text, boxes, region labels	Inspect or replace each stage	Missed text; wrong reading order
Text + layout	Tokens and geometry	Model spatial relations	OCR and coordinate errors
Direct visual	Page or crop embeddings	Retain visual appearance	Fine detail lost in resizing or compression
Hybrid / tools	Text, crops and linked outputs	Re-observe difficult regions	Misaligned or conflicting outputs

Table 1 Perception routes compared by their evidence interface. These are design trade-offs, not a ranking of benchmark scores.

Table 1 summarizes the resulting choices. For the financial example in Figure 2, all routes must recover both the table and the restatement note. OCR provides inspectable strings and cells; direct visual input preserves appearance; a hybrid route can combine the two. Their shared requirement is the correct table–footnote link, not a particular model family.

2.2.4 Evaluate the Output That the Task Uses

Transcription, parsing and answering require different diagnostics. Character or word error measures reading fidelity, but does not test whether a table cell is attached to the right header. Parsing benchmarks inspect page structure and heterogeneous elements; answer metrics inspect the downstream result. A system may improve one while leaving another unchanged.

OCRBench aggregates 29 datasets spanning recognition and several text-dependent visual tasks [42]. OCRBench v2 places greater emphasis on localization and reasoning [18], while OmniDocBench provides detailed annotations for diverse PDF parsing [54]. Their scores answer different questions and should be reported with input resolution, page access and output requirements.

For document QA, the most useful error analysis follows a value back to its source: was the character read correctly, assigned to the right region, connected to the right label and used in the intended calculation? This separates a recognizer improvement from a structural or reasoning improvement. It also identifies when another crop or a specialist tool is warranted, instead of escalating every page to a more complex pipeline.

3 Table Understanding and Question Answering

A table answer depends on relations: which header governs a cell, which unit applies, and which rows satisfy the question. Table QA methods differ in where they represent and transform these relations. Figure 5 follows the evidence from a source table to an answer; Table 2 locates representative methods along that workflow.

3.1 Preserving Rows, Columns and Units

Flattening a table into prose can separate a value from its header or merge unrelated columns. Structure-aware pre-training instead teaches models to associate cell content with row, column and contextual information. TAPAS-style intermediate pre-training [16] and OmniTab [29] develop table–question alignment through tabular learning objectives. FORTAP [9] uses spreadsheet formulas to connect cell relations with numerical operations. These approaches improve the representation supplied to a reasoning model; they do not replace the initial recovery of structure from a document image.

Representation remains important after pre-training. Structural-noise analysis [60] shows why changing serialization or disrupting headers can alter an LLM's interpretation of the same cells. A useful input therefore retains header scope, units, missing-value conventions and source coordinates. A clean CSV already supplies much of this information; a scanned financial table requires a separate extraction step. Evaluating the two settings as the same task conceals an important source of error.

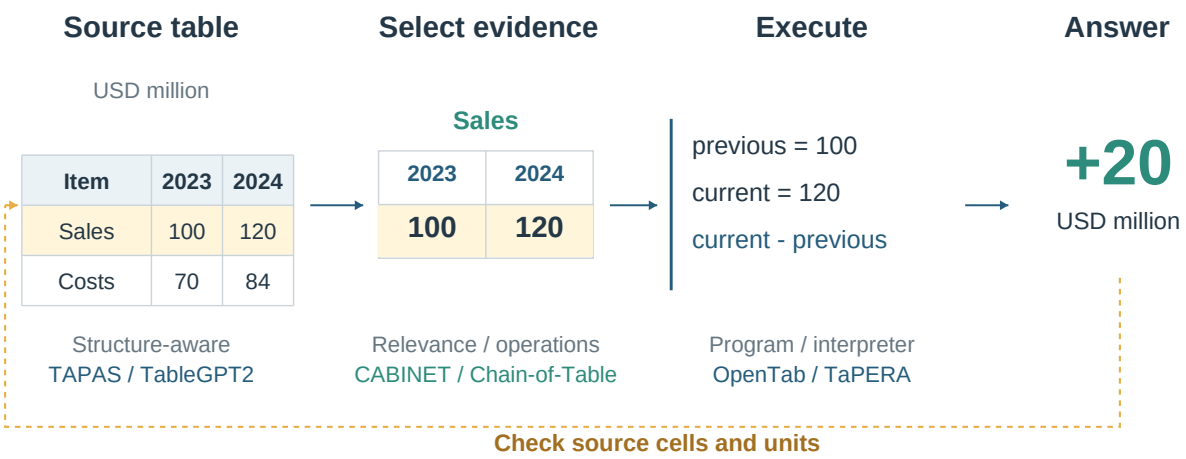


Figure 5 From a source table to a computed answer. Selection, table transformations and executable programs expose different intermediate objects for inspection.

3.2 Selecting and Transforming Evidence

Large tables create a second problem: the relevant evidence may occupy only a few rows. CABINET [55] uses question-conditioned relevance weighting to suppress distracting content. Selection reduces context, but it must retain the headers and comparison values needed to interpret the remaining cells. Removing an apparently irrelevant unit or denominator can make a compact sub-table unusable.

Other methods transform the evidence rather than only selecting it. Chain-of-Table [75] maintains a table as intermediate state and updates it through operations such as filtering and aggregation. DePlot [41] first translates a chart into a table, allowing a language model to reason over extracted values. Their intermediate objects make different errors visible: an incorrect operation in the former, an incorrect visual-to-tabular conversion in the latter.

Instruction tuning changes how a model responds to tabular tasks. Table-GPT [36] uses synthetic table-task data, while TableGPT2 [62] combines a specialized table encoder with a language decoder. TableGPT2's additional modality is schema- and cell-level tabular information; it should not be confused with directly reading a table image. Such models can also be combined with selection, retrieval or execution rather than treated as alternatives to those mechanisms.

3.3 Computing and Checking the Answer

Numerical questions benefit from explicit computation. OpenTab [31] combines table retrieval with generated SQL. TaPERA [92] uses content planning and Python execution for long-form table answers; TAT-LLM [96] addresses discrete reasoning over tables and text. These designs expose operands and intermediate results, making arithmetic easier to inspect than a verbal explanation alone.

Execution nevertheless checks a narrower property than answering the question. A valid SQL query can select the wrong accounting period, and correct subtraction can use an unadjusted value. For a year-on-year margin question, the workflow must connect each value to its year, preserve the percentage unit, apply any restatement and then compute the change. The source table, selected cells, executed operation and final unit should remain linked. The cross-page example in Figure 2 illustrates this dependency.

3.4 Choosing a Table QA Workflow

The input and question should determine the workflow. For a small, clean table, a direct reader with arithmetic execution may suffice. For a wide table, select a compact sub-table while retaining its schema. For an open collection, retrieve the table before reasoning over its cells. For image-based evidence, inspect extraction

Method	Main mechanism	Inspectable object	Essential source check
TableGPT2 [62]	Schema/cell encoding	Table and response	Relevant cells support answer
CABINET [55]	Relevance weighting	Selected content	Required headers survive
Chain-of-Table [75]	Iterative table operations	Intermediate tables	Operations match question
DePlot [41]	Chart-to-table conversion	Extracted values	Values match chart
OpenTab [31]	Retrieval and SQL	Table, SQL, result	Correct table and columns
TaPERA [92]	Planning and Python	Plan, program, output	Inputs and plan match source

Table 2 Table QA methods compared by the evidence object they expose. The final column identifies checks that remain beyond successful generation or execution.

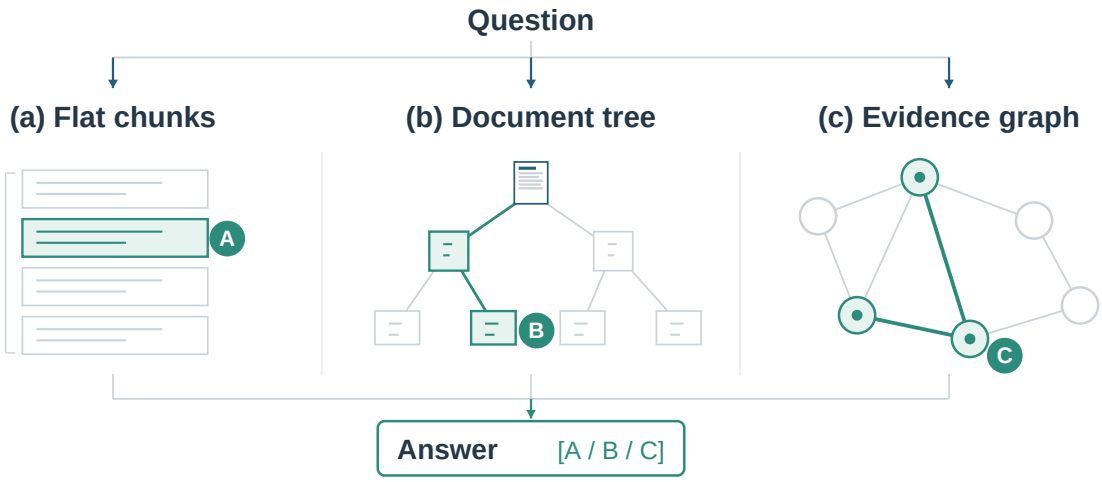


Figure 6 Three ways to organize textual evidence for retrieval: flat passages, document hierarchy and linked entities. The index representation and the policy for making additional retrieval calls are separate choices.

quality before attributing an error to downstream reasoning. Table 2 supports this choice without collapsing representation, selection and execution into a single model ranking.

4 Text-Based Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) gives a generator access to an external information store [33]. In document QA, the central design choice is how to divide and reconnect evidence: a question may need a passage, its section context and a reference on another page. Figure 6 contrasts flat, hierarchical and graph-based access to the same document collection.

4.1 What Does the Index Preserve?

A flat passage index is simple to build and update. Each chunk receives a retrieval representation, and the most relevant chunks are passed to the generator. Its weakness is the boundary: a table caption, paragraph or qualification can fall outside the selected chunk. Increasing overlap reduces some boundary losses but repeats content and consumes context. Source document, section and page identifiers are therefore useful even when retrieval itself operates over plain text.

Hierarchical indexes preserve containment. PdfTriage [59] exposes document structure for questions referring to pages, sections or tables. HiQA [6] augments retrieval with hierarchical context, while discourse-aware retrieval [47] uses rhetorical structure to organize long-document evidence. The common operation is to retrieve a local match together with enough parent or neighboring context to interpret it. This helps distinguish similarly worded passages that belong to different products, years or experiments.

Graph indexes preserve relations across boundaries. GraphRAG [15] constructs entity–relation graphs and community summaries for global, cross-document questions. LightRAG [23] retrieves entity and relational information at two levels; HyperGraphRAG [44] represents relations involving several entities with hyperedges. These structures can expose evidence paths that passage similarity misses. Their usefulness depends on the extracted links: a false entity merge can connect unrelated facts, while an omitted relation can break a required path.

The three representations can coexist. A graph can nominate documents, a hierarchy can locate a section and a passage index can retrieve the final quotation. BookRAG [72] combines hierarchical and relational organization in this spirit. The practical question is which relation the task requires—containment, adjacency, entity identity or a cross-reference—and whether the indexing process preserves it.

4.2 When Should Retrieval Continue?

Index structure does not determine runtime control. A graph can be queried once, and a flat index can support repeated search. One-pass retrieval suits questions whose evidence need is clear at the outset. Iteration is useful when an initial passage reveals a new entity, a missing comparison or a reference that must be followed. Triple-Fact Retriever [80] connects evidence hop by hop; DR-RAG [26] adapts relevance selection, and GARLIC [74] controls access through a hierarchical weighted graph.

Every additional action should address an observable gap. A useful trace records the current question, the new source found and the unresolved requirement. A stopping rule can then distinguish sufficient evidence from a search that merely repeats similar passages. Without such control, query drift or an early mistaken entity can direct later retrieval toward a coherent but irrelevant set of sources. Longer traces should consequently be compared with a fixed-pass baseline under the same call and context budget.

4.3 Retrieval or Full Context?

Full-context processing avoids a separate retrieval recall failure when the document fits, but the reader must still find and combine the relevant information. Retrieval is attractive for large, changing collections when each question needs only a small subset. LaRA [34] examines this trade-off rather than assuming one route always wins. Summarization, as in MemSum-DQA [21], offers another compression point: it reduces input length but may discard details needed by a later question.

Comparison should therefore separate three failures: the evidence never reached the reader, the reader selected the wrong part, or the answer misused available evidence. An oracle-evidence run isolates the reader; an end-to-end run includes retrieval. Report both when possible, together with the number of retrieved units, input tokens and retrieval calls. Keeping source identity through each step connects text RAG with table and visual document QA: compression is useful only if the information needed to support the answer survives.

5 Vision-Language Models for Document Understanding

A visual document reader must solve two problems: preserve enough page detail to read the evidence, and expose enough pages to answer the question. These compete for the same image and context budgets. Model size alone does not resolve the trade-off. We organize this section around representation, visual resolution, retrieval granularity and control, using the concrete mechanisms in Table 3.

5.1 Representation: Text, Coordinates and Pixels

Layout-aware encoders preserve explicit word positions; LayoutXLM extends this approach to multilingual documents [83]. UDOP instead combines text, visual appearance and layout within a generative document processor [65]. These interfaces should be distinguished from DocLLM, which generates from text and bounding boxes without an image encoder [68]. A layout-aware input can retain a column association that plain text loses, but coordinates alone cannot describe a diagram that was never extracted.

Direct-visual readers pass page images to the model. Donut generates structured outputs without a separate OCR engine, while Pix2Struct learns image-to-text parsing through screenshot-based pre-training [30, 32].

Route	Input	Evidence interface	Key constraint
DocLLM [68]	Text + boxes	Spatial attention	OCR / layout quality
Donut [30]	Page image	Structured generation	Small-text readability
mPLUG-DocOwl2 [28]	Page images	Compressed visual tokens	Detail versus page coverage
ColPali [17]	Query + page corpus	Ranked multi-vector pages	Index size; retrieval recall
VisRAG [86]	Retrieved images	Selected visual context	Missing evidence pages
DocLens / ViDoRAG [71, 95]	Adaptive page access	Inspection / retrieval actions	Calls; stopping decisions

Table 3 Visual-document interfaces expose different evidence and bottlenecks. Representation, retrieval and adaptive control can be combined within one system.

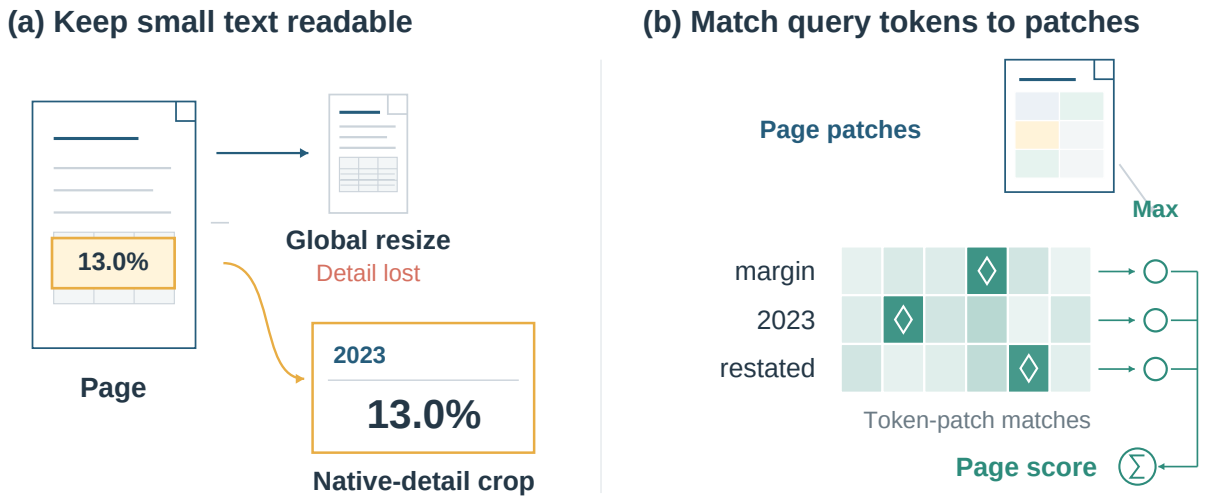


Figure 7 Two visual-budget decisions: preserve small-text detail through crops, and preserve query-specific matches through patch-level late interaction. The matching matrix is schematic, not a measured retrieval result.

This removes an external transcription stage, rather than the task of reading. Small print, mathematical symbols and relations between distant regions still need to survive image encoding. The practical comparison is therefore between the evidence accessible through each interface, not between an "OCR" and "OCR-free" label.

The output interface also matters. A document converter must recover content and structure across a page; a question-answering model may only need one value. SmolDocling expresses document elements through DocTags, while general readers such as Qwen2.5-VL and Qwen3-VL support broader visual-language tasks [1, 2, 51]. A correct answer to one query does not show that the underlying page has been converted completely. Conversely, a complete transcription does not establish that the right evidence was selected for a question.

5.2 Visual Budget: Readable Detail versus Page Coverage

Figure 7 illustrates the central resource trade-off. Shrinking a page preserves its overall arrangement but can erase the characters needed to read a footnote. Cropping preserves local detail but requires additional visual tokens and a way to reconnect regions with their source page. A document reader therefore needs both an effective resolution and a policy for allocating that resolution.

For a patch encoder, a simple budget model is

$$N_{\text{visual}} \approx \sum_{p=1}^{P} \sum_{c=1}^{C_p} n_{p,c},$$

where P is the number of supplied pages, C_p the crops retained from page p, and $n_{p,c}$ the tokens for each crop after compression. This accounting separates three choices often hidden behind a maximum-context specification: how many pages are supplied, how finely each is sampled, and how much information the model compresses. It is a resource description, not a performance law.

Monkey and Qwen2-VL study high-resolution or variable-resolution visual processing [39, 70]. Structure learning in mPLUG-DocOwl 1.5 and visual slimming in DocKylin address complementary parts of the interface [27, 90]. More pixels can make a character readable; a better representation can make its relation to a header usable. Neither repairs an omitted page. A useful resolution experiment should therefore separate local reading errors from errors caused by missing context.

Coarse-to-fine document conversion offers another allocation strategy: inspect a low-resolution global view, then read selected regions at higher resolution. MinerU2.5 uses this decoupling for document parsing [53]. The benefit is selective expenditure of visual computation; the risk is that a region rejected during the coarse pass never receives a detailed reading. Region coverage and reading accuracy should thus be examined together, particularly for small notes and irregular tables.

Multi-page readers add a further compression boundary. mPLUG-DocOwl2 reduces visual-token pressure for multiple images, while Docopilot studies document-level processing as a direct-reading route [14, 28]. Their design question differs from single-page OCR: which page details remain distinguishable when many pages share one context? Tests should hold document access constant while varying page count and resolution, rather than supplying one model with a complete report and another with isolated crops.

The original MMLongBench-Doc study illustrates why this distinction matters: 12 of its 14 evaluated LVLMs performed worse than corresponding LLMs given OCR-parsed text [46]. The finding belongs to that study's model versions and budgets. Its broader methodological lesson is to include a strong text-based reader when extracted text contains the required evidence, and to isolate questions that actually depend on visual structure before attributing gains to direct image access.

5.3 Retrieval: Pages, Regions and Cross-Page Evidence

Visual retrieval indexes page appearance, retaining diagrams and spatial cues that a text-only index may omit. ColPali represents a page with multiple vectors and uses late interaction with query tokens [17]. In simplified form, its scoring principle is

$$s(q, d) = \sum_{i} \max_{j} \langle q_i, d_j \rangle,$$

where each query-token vector q_i matches its most compatible page vector d_j. Unlike a single pooled vector, this retains several query-specific matches, with additional index storage and scoring cost.

Document screenshot embeddings offer a unified image-based retrieval input, while VisRAG connects visual retrieval to answer generation [45, 86]. The retrieved object is still an evidence candidate. A relevant page can contain a distracting table, and several individually relevant pages can omit the one footnote needed for a comparison. Retrieval must therefore be assessed both as ranking and as coverage of the complete evidence requirement.

The retrieval unit changes that balance. Page retrieval preserves surrounding context but sends irrelevant content to the reader. RegionRAG selects finer regions; hierarchical indexing in MHier-RAG provides an intermediate organization across document scales [20, 37]. Smaller units improve focus only if headers, units and cross-references survive. For a table cell, the useful evidence unit may be the cell together with its row label, year header and qualifying note, rather than its bounding box alone.

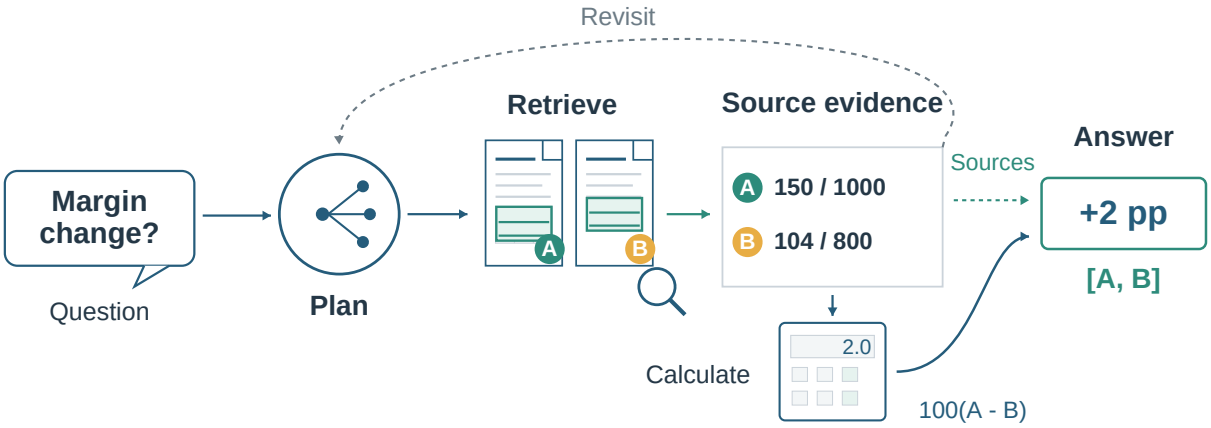


Figure 8 Adaptive document access links an intermediate evidence need to the next inspection or tool action. The loop ends with supported evidence, an exhausted budget or abstention.

Cross-document systems add evidence composition. M3DocRAG studies visual retrieval over multiple pages and documents, and RAG-Anything organizes heterogeneous content through graph structures [10, 24]. These mechanisms are useful when a question crosses document boundaries or modalities. Their value should be tested against the simpler alternative of supplying the complete relevant document: retrieval saves context only when the savings outweigh omitted or disconnected evidence.

5.4 Adaptive Access and Stopping

An adaptive controller chooses what to inspect next after seeing an intermediate result. Figure 8 shows this feedback loop. A search can identify an entity; the next action can open its table, inspect a small note, or execute a calculation. This differs from retrieving a fixed set of pages once, but can use the same underlying OCR, visual encoder and index.

DocLens provides tools for inspecting document evidence, while ViDoRAG combines multimodal retrieval with iterative exploration [71, 95]. MA-RAG investigates coordinated agents, and CogDoc uses coarse-to-fine reasoning [52, 81]. The shared mechanism is conditional access: later evidence depends on earlier findings. Dividing the same fixed context among several agents is a different intervention and should be compared separately.

A controller also decides how much work a query deserves. AutoThinkRAG explores query-dependent routing of retrieval and reasoning [84]. A direct lookup may need one page; a cross-document comparison may need repeated access. Adaptive effort is most useful when these needs vary across queries. Reporting the average number of calls alone can hide a policy that spends heavily on easy questions and stops early on the difficult ones.

A practical stopping decision asks whether the required evidence has been found and whether the requested operation can be checked. Repeated searches without new support add latency and opportunities for drift. For a small, complete document, a fixed-context reader remains an essential control; for a large collection, fixed retrieval supplies a useful efficiency baseline. Compare these routes under matched page, token and tool budgets, and retain page or region identifiers for the final claims.

6 Evaluation

A useful evaluation locates the failure: unreadable input, missing evidence, unsupported interpretation or incorrect computation. One answer score cannot distinguish these causes. We organize benchmarks by the evidence they expose, then separate four observable properties of a document-QA response.

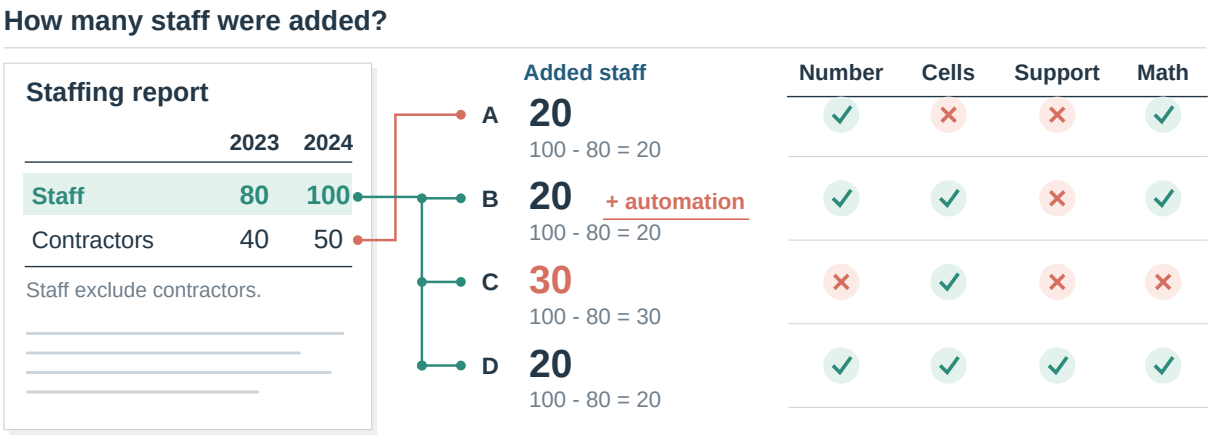


Figure 9 Constructed response traces separate numeric agreement, evidence localization, source support and arithmetic. The examples illustrate diagnostic outcomes, not measured model scores.

6.1 Four Checks on One Answer

Figure 9 contrasts responses to the same source. A correct number can accompany the wrong citation; a relevant citation can accompany an invented explanation; and the correct cells can still be combined with faulty arithmetic. Evaluation should preserve these distinctions rather than collapse every response into a single success flag.

Answer correctness concerns the requested result and its unit. Exact match tests normalized string identity; token F1 measures overlap; ANLS uses normalized edit distance with the benchmark's threshold. These metrics suit different answer forms and should not be averaged across datasets as interchangeable percentages. For open-ended answers, the judge model, prompt and a human-audited sample help reveal whether fluent paraphrases or unsupported additions are being rewarded.

Evidence localization concerns the selected page, passage, region or cell. Recall measures whether required evidence is present; ranking metrics such as nDCG and MRR also consider its position. Gold-page experiments isolate the reader, whereas end-to-end retrieval experiments include evidence selection. Both are useful, but answer scores obtained with gold evidence should be reported separately from scores that require searching a collection.

Source support asks whether the cited material justifies the complete answer, including qualifications and explanations. A correct numerical component does not support an additional causal claim. Equivalent evidence may occur in several locations, so annotation should allow valid alternatives. Support is also distinct from causal faithfulness: establishing whether the model actually relied on the cited source requires interventions such as removing or substituting that evidence, not just checking the displayed citation.

Execution correctness asks whether the declared operation ran correctly on the supplied inputs. A Python or SQL result can verify arithmetic while using the wrong year, unit or table. It should therefore be paired with localization and semantic checks on those inputs. For multi-step questions, retain the intermediate values and their source locations so that a computational error can be separated from a parsing or planning error.

6.2 Choose Benchmarks by Capability

Table 4 provides a compact selection map. The rows differ in task scope, evidence annotation and answerability; they do not form a common leaderboard. A page-reading benchmark is a weak test of collection search, while a retrieval benchmark alone does not test whether the final explanation is supported.

For single-page reading, DocVQA emphasizes document text and layout, while InfographicVQA adds graphics

Resource	Scope	Released supervision	Main metrics
DocVQA [48]	One page	Answer strings	ANLS
InfographicVQA [49]	Infographic	Answers; categories	ANLS
QASPER [12]	Scientific paper	Answers + evidence; unanswerable	Answer / Evidence F1
SlideVQA [63]	Slide deck	Answers + pages + expressions	QA / evidence EM, F1
DUDE [67]	Multiple pages	Multiple answer types; unanswerable	ANLS-based scoring
MMLongBench-Doc [46]	Long visual PDF	Answers + source pages; unanswerable	Answer F1
ViDoRe (original) [17]	Page retrieval	Relevant-page judgments	nDCG@5; Recall@K
CRAG [85]	Factual RAG	Answers; invalid queries; graded responses	Accuracy; hallucination

Table 4 Eight evaluation resources selected by task and released supervision. Evidence annotations are not necessarily provided to the tested reader; input access must be stated.

and visual organization. The latter's first preprint appeared in 2021, before its WACV 2022 publication. Multi-page benchmarks expose a different difficulty: QASPER links answers to supporting paper content, and SlideVQA includes evidence pages and numerical expressions. These annotations enable diagnostics that answer strings alone cannot support, including whether the system found all necessary pages.

Long documents and collections separate context capacity from search quality. MMLongBench-Doc tests mixed visual evidence across long PDFs; ViDoRe tests query-to-page retrieval. Evaluating both can reveal whether a visual RAG pipeline fails before or after evidence reaches its reader. MultiHop-RAG adds another axis: composing support across more than one source [64]. A high single-page score provides little information about that composition step.

Domain and document quality should be selected deliberately. LegalBench-RAG focuses retrieval on legal material, while REAL-MM-RAG studies heterogeneous enterprise evidence [58, 76]. M6Doc and RoDLA instead probe layout diversity and robustness [7, 8]. These resources answer different questions: domain retrieval can fail despite clean OCR, and degraded page structure can fail before retrieval begins. A practical evaluation combines the relevant stages rather than treating one benchmark as a universal proxy.

Answerability and freshness require separate cases. DUDE and QASPER include unanswerable questions; CRAG distinguishes accurate, hallucinated and missing responses. Dedicated unanswerability studies examine when a RAG system should decline [56]. FreshStack evaluates retrieval in a changing technical corpus [66]. Report refusal together with coverage and error rate: a system that declines every difficult question can appear conservative while supplying little useful information.

6.3 A Matched Comparison Protocol

Match the document snapshot and split; raw images versus native text or supplied OCR; gold pages versus collection search; image resolution, crops and context budgets; model or API version; retrieval depth and tool limits. Include full-context and OCR-based controls where they receive equivalent information. Report answer and evidence metrics alongside ingestion cost, online latency and failed requests, with breakdowns for document length, scan quality, language and evidence modality. This protocol makes the comparison interpretable without turning every architectural change into a claim of general superiority.

7 Research Priorities and Conclusion

Three research priorities follow from the interfaces examined in this survey.

Preserve evidence across conversions. Text, coordinates, cells and image patches should retain a reversible link to their source page. In the running example, a number without its year or restatement is insufficient.

Future systems should expose what is discarded during serialization, cropping and token compression, and test whether an answer changes when this information is restored.

Spend computation where evidence is missing. A clean table may need only a reader and calculator; a small footnote may need a high-resolution crop; a cross-document question may need retrieval. Compare fixed context, one-shot retrieval and iterative access under the same source and inference budget. Additional tool calls are useful when they recover evidence or correct an error, rather than repeat a plan.

Evaluate the complete support chain. Measure answer accuracy, evidence localization, source support and execution separately. Include cases with missing evidence, distracting pages, degraded scans and changing document collections. Evidence-removal or substitution controls can test reliance on a claimed source, while accuracy–coverage curves reveal whether abstention is useful.

Document intelligence therefore requires more than fluent answers or accurate OCR in isolation. The practical design question is which representation and access pattern preserve the evidence required by a task. The visual workflows and mechanism comparisons in this survey make that question explicit across perception, retrieval, table computation and multimodal reasoning.

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